

Raw Data Entry Worksheet

Data Entry: All Data Entry for Form LA-ANR and LA-CP₂ (Orange Tabs) will be entered on this worksheet. Data entered on this worksheet will automatically populate the LA-ANR and LA-CPL forms.

General Information

Facility Name: Cadillac WWTF
 City: Cadillac MI
 Reporting Year: 2019
 NPDES No.: MIO020257

Previous Year*	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8	Site 9	Site 10
Next Year										

*If you DID NOT apply sludge in the previous year DO NOT enter data, please leave the cell BLANK.

Site Number	Site Name	Site Location	Application Period		Sludge Applied*		Percent Solids of Sludge Applied	Acres Applied / Site	Sludge App Rate DRY TONS / ACRE**
Site No. 1	W32-NS01	32-23N-10W	6/30/2019	7/1/2019	555,262		4.32	42.00	2.38
Site No. 2	W29-NS01	29-23N-10W	6/26/2019	6/29/2019	191,600		4.76	27.00	1.41
Site No. 3	W20-TC01	20-21N-9W	10/16/2018	10/16/2019	110,308		2.08	17.00	0.56
Site No. 4	W19-TC01	19-21N-9W	10/16/2019	10/17/2019	45,562		2.12	8.00	0.50
Site No. 5	W17-EK01	17-21N-9W	10/17/2019	10/19/2019	201,495		2.43	15.00	1.36
Site No. 6	W17-RW01	17-21N-9W	10/17/2019	10/19/2019	201,495		2.43	19.00	1.07
Site No. 7	W22-JR01	22-21N-9W	10/19/2019	10/19/2019	109,598		2.5	23.00	0.58
Site No. 8	W27-JR01	27-21N-9W	10/20/2019	10/20/2019	64,746		2.76	14.00	0.53
Site No. 9									
Site No. 10									

*If you applied sludge in gallons ONLY list the gallons applied. DO NOT calculate the pounds applied.

**Program will calculate these values.

Heavy Metals Calculations

If the result is less than the analytical detection limits enter the less than sign (<) followed by the minimum reportable limit but DO NOT enter acronym's such as, "ND", "BPQL", "BMRL", etc.

[illegible]

For Calculating the Agronomic Rate for the Land Application of Sewage Sludge

Step 1:

From analysis of the sewage sludge to be land applied determine the amount of each nitrogen compound, based on dry weight, in pounds per ton (lbs/ton).

Nitrogen compound	Concentration of Nitrogen Compounds							
	Sewage (mg/kg)	Sewage (mg/kg)	Sewage (mg/kg)	Sewage (mg/kg)	Sewage (mg/kg)	Sewage (mg/kg)	Sewage (mg/kg)	Sewage (mg/kg)
Total Kjeldahl Nitrogen (TKN as N)	94,400	94,400	57,000	57,000	57,000	57,000	57,000	
Ammonium Nitrogen (NH_4 as N)	4,500	4,500	22,300	22,300	22,300	22,300	22,300	
Nitrate Nitrogen (NO_3 as N)	215	215	30	30	30	30	30	
Total Phosphorus (mg/kg or mg/L)	39,300	39,300	48,200	48,200	48,200	48,200	48,200	

Step 2:

Assume that the available friction (K_v) is dependent upon operations at the site (see Table 1). Use the following table and enter the appropriate K_v Factor

If Sewage Sludge is:	Factor K_d is:
Liquid and Surface Applied	0.25
Liquid and Incorporated into the Soil	1.00
Devalerized and Applied in any Manner	1.00

Factor K _i 's	1	1	1	1	1	1	1	1	

Step 3:

Calculate the amount of organic nitrogen available in the sewage sludge to be applied. The factor f , used for determining the amount of Org as N present due to mineralization, is provided below in Table 2. The value of F is dependent upon how the sludge is treated (i.e. aerobic digestion composted etc.)

F Values				
Time After Sludge Application (Year)	Stabilized Primary and Waste Activated Sewage Sludges, Fraction of Org as N	Aerobically Digested Sewage Sludge Fraction of Org as N	Anaerobically Digested Sewage Sludge Fraction of Org as N	Composted Sewage Sludge Fraction of Org as N
0.10 to 1	0.40	0.30	0.20	0.10

F Values	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20
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Step 4

Calculation is determined on data entered in Steps 1 through Steps 3

Nitrogen Credits. The available nitrogen credits from previous legume crops and existing residuals must be determined under there will be a previous crop nitrogen credit in the soil. Select the appropriate nitrogen credit based upon the Step 5a. Previous Legume Crop. Previous Crop. If the crop for the previous year was a legume and was plowed data shown in Table 3.

Table 3. Nitrogen Credits from Legumes in Rotations	
Legume Crop	Nitrogen Credits
Alfalfa 1st year after: >50% stand 60-80% stand <50% stand	100-140 lbs/acre 60-100 lbs/acre 0-60 lbs/acre
Alfalfa 2nd year after)	Half of 1st year credit
Sweet Clover Red Clover	100-120 lbs/acre 40-80 lbs/acre
Soybeans*	30-60 lbs/acre

*Yield: 1 pound of N credit per bushel of pods
 *Not credited for wheat double-cropped after soybean harvest.)

If you do not have a "nitrogen credit" please enter 0 for the appropriate site

[illegible]

Step 5B:

The nitrogen credit for the existing nitrate level in the soil can be accounted for by using the soil test nitrate results in the following equations. (See soil testing procedures for soil sampling methods)

[illegible]

Step 5C:

Previous Sludge: Available Organic Nitrogen: Previous OrgN/Available. Because the mineralization of organic nitrogen in sludge occurs over a long time, there will be a nitrogen credit for mineralization of previously applied sludge. This nitrogen credit is 0.5 of the previous year's calculated total available organic nitrogen.

Abbreviations	Definition	Location of Information
PYCONA	Previous Year Current Organic (lb/ston) N_{pycon}	Step 3 of Last Years Report
PYSAR	Previous Years Sludge Application Rate (Dry Tons / Acre)	Last Year Data Entry - General Information Section

[illegible]